

Analysis Trophic Network Noronha

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Overview

This report computes the total (direct + indirect) effects among species in the Fernando de Noronha food web using dietary interaction data. The workflow:

1. Loads dietary and species data.
2. Builds the interaction matrix.
3. Computes the total effects matrix.
4. Summarises positive and negative effects.
5. Visualises the interaction network.
6. Performs a sensitivity analysis for the interaction dependency parameter R .

Load packages

```
Require::Require("data.table")
Require::Require("reshape")
Require::Require("ggplot2")
Require::Require("cowplot")
```

Import data

```
psiri <- as.matrix(read.csv("data/psiri.csv", header = TRUE, row.names = 1))
rawFreq <- as.matrix(read.csv("data/rawFreq.csv", header = TRUE, row.names = 1))
spp <- read.csv("data/species.csv", header = TRUE)
```

Reorder species

```
ord <- c(
  'r1', 'r2', 'r3', 'r4', 'r5', 'r6',
  'm1', 'm2', 'p1', 'p2', 'p3'
)
```

```

psiri <- psiri[ord, ord]
rawFreq <- rawFreq[ord, ord]

spp <- as.data.table(spp)
spp[, abr := factor(abr, levels = ord)]
setkey(spp, "abr")

lookup <- setNames(spp$name, as.character(spp$abr))

rownames(psiri) <- rownames(rawFreq) <- lookup[rownames(psiri)]
colnames(psiri) <- colnames(rawFreq) <- lookup[colnames(psiri)]

```

Build the interaction matrix

The four invasive predators are weighted according to their estimated densities after correcting dietary frequencies by sample size.

```

Bw <- rawFreq

samplesInvaders <- c(
  "felis_catus" = 78,
  "salvator_merianae" = 22,
  "rattus_rattus" = 10,
  "rhinella" = 66
)

Bw[, names(samplesInvaders)] <-
  Bw[, names(samplesInvaders)] /
  matrix(
    samplesInvaders,
    ncol = 4,
    nrow = nrow(rawFreq),
    byrow = TRUE
  )

densities <- c(
  "felis_catus" = 0.71,
  "salvator_merianae" = 3.98,
  "rattus_rattus" = 37,
  "rhinella" = 10.35
)

Bw[, names(samplesInvaders)] <-
  Bw[, names(samplesInvaders)] *
  matrix(
    densities,
    ncol = 4,
    nrow = nrow(rawFreq),
    byrow = TRUE
  )

```

```
M <- 0.1 * t(psiri)
M <- M + Bw
```

Species interaction dependency

```
R.vec <- c(
  invertebrados_ter      = 0.1,
  invertebrados_aq      = 0.1,
  rodentia               = 0.5,
  kerodon_rupestris     = 0.5,
  birds                 = 0.1,
  amphisbaena_ridleyi   = 0.2,
  trachylepis_atlantica = 0.5,
  rattus_rattus          = 0.8,
  felis_catus           = 0.8,
  salvator_merianae     = 0.8,
  rhinella              = 0.8
)

stopifnot(setequal(names(R.vec), colnames(M)))
R.vec <- R.vec[colnames(M)]
```

Function to compute total effects

```
indirect <- function(mat, R) {

  stopifnot(is.matrix(mat))
  stopifnot(nrow(mat) == ncol(mat))

  stopifnot(is.numeric(R))
  stopifnot(length(R) == nrow(mat))

  stopifnot(!anyNA(mat))
  stopifnot(!anyNA(R))

  S <- nrow(mat)

  row_totals <- rowSums(mat)
  stopifnot(all(row_totals > 0))

  W <- mat / row_totals

  consumers <- spp$name[spp$is_consumer]
  prey <- setdiff(rownames(W), consumers)

  W[prey, consumers] <- -W[prey, consumers]

  P <- diag(R)
```

```

I <- diag(S)

Tmat <- solve(I - P %*% W)

T_no_diag <- Tmat
diag(T_no_diag) <- NA

mean_contribution <- colMeans(T_no_diag, na.rm = TRUE)

B <- (mat > 0) * 1

indirect_only <- T_no_diag * (1 - B)

prop_ind_eff <-
  sum(indirect_only, na.rm = TRUE) /
  sum(T_no_diag, na.rm = TRUE)

prop_ind_eff_tin <-
  rowSums(indirect_only, na.rm = TRUE) /
  rowSums(T_no_diag, na.rm = TRUE)

prop_ind_eff_tout <-
  colSums(indirect_only, na.rm = TRUE) /
  colSums(T_no_diag, na.rm = TRUE)

list(
  mean_contribution = mean_contribution,
  prop_ind_eff = prop_ind_eff,
  prop_ind_eff_tin = prop_ind_eff_tin,
  prop_ind_eff_tout = prop_ind_eff_tout,
  direct = P %*% W,
  Tmat = Tmat
)
}

```

Compute total effects

```

T.matrix <- indirect(M, R.vec)

Tm <- T.matrix$Tmat

```

Positive and negative contributions

```

NT <- pmin(Tm, 0)
NTout <- colSums(NT)

PT <- (Tm > 0) * Tm
PTout <- colSums(PT)

```

```
sort(NTout)
```

```
## [1] -0.93189674 -0.41114653 -0.39922503 -0.26514712 -0.23548766 -0.18793515  
## [7] -0.10837706 -0.08734012 -0.07751703 -0.06645093 -0.05894146
```

```
sort(PTout, decreasing = TRUE)
```

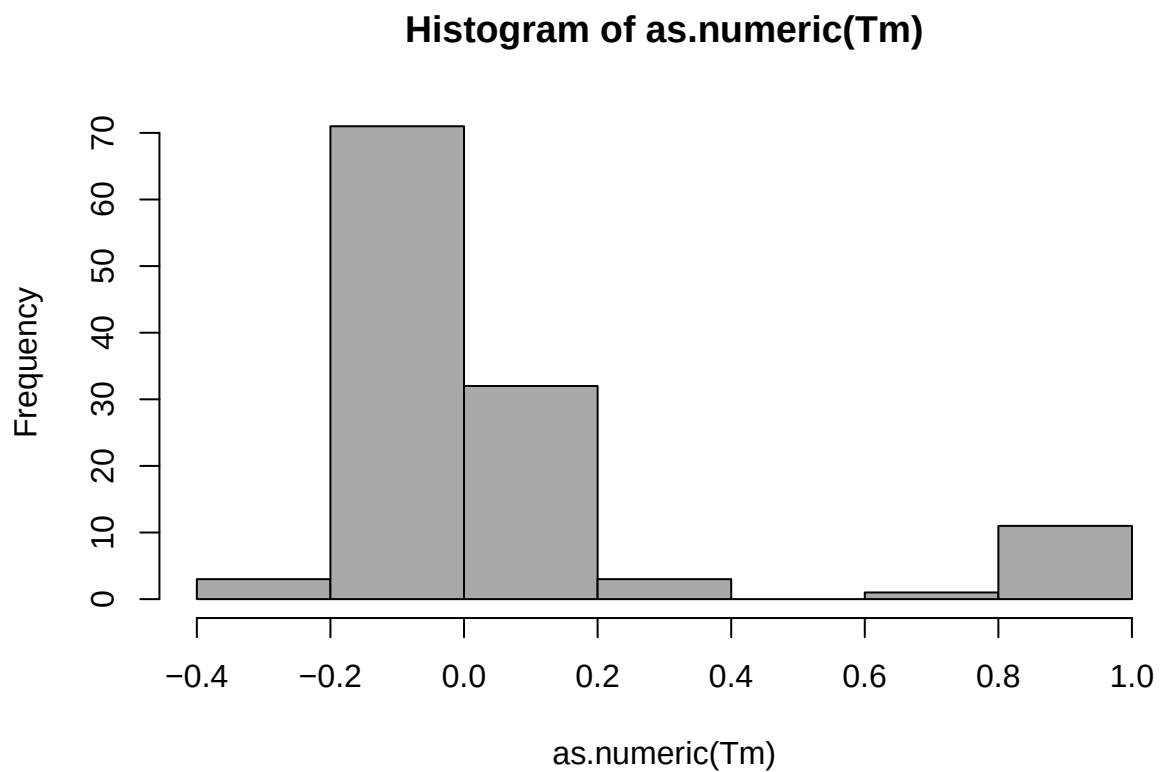
```
## [1] 2.0160112 1.5366731 1.2395869 1.1506342 1.1270741 1.1086980 1.0665436  
## [8] 0.9609929 0.9602916 0.9329409 0.8446811
```

```
mean(NT)
```

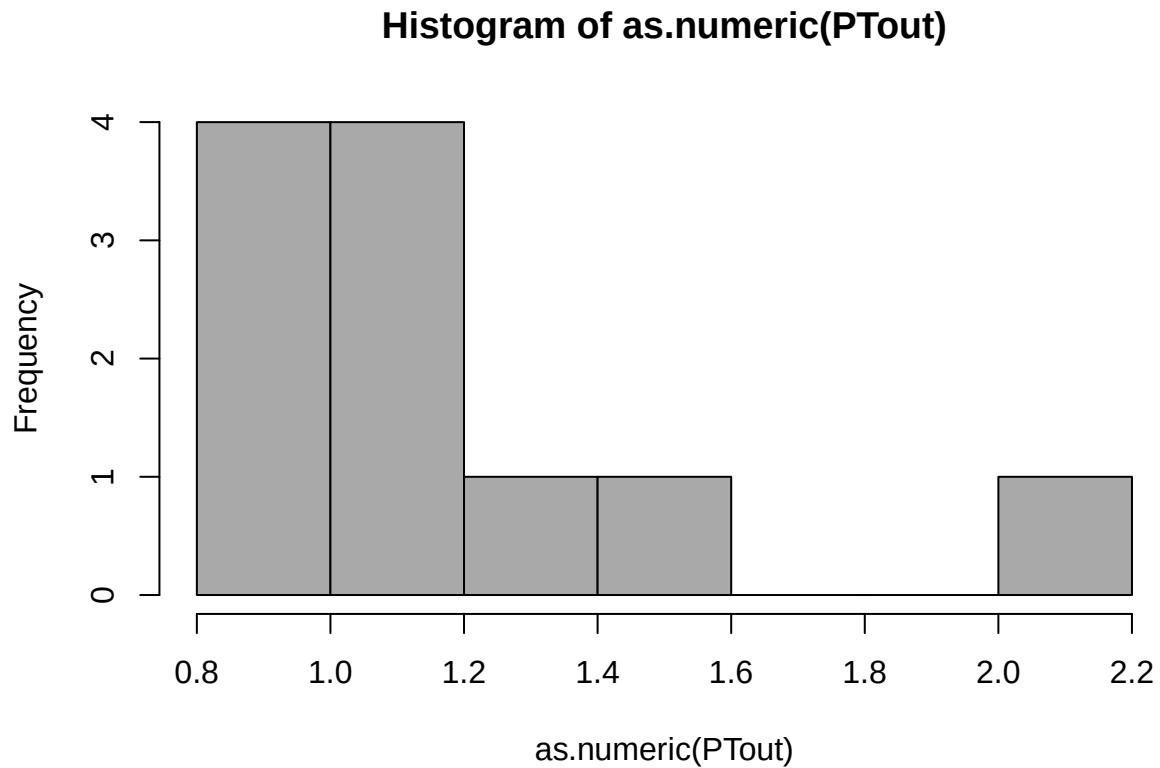
```
## [1] -0.02338401
```

Distribution of total effects

```
hist(as.numeric(Tm), col = "darkgrey")
```



```
hist(as.numeric(PTout), col = "darkgrey")
```



Heat map of total effects

```
Tm_plot <- Tm
diag(Tm_plot) <- NA

df <- as.data.table(as.table(Tm_plot))
colnames(df) <- c("x", "y", "value")

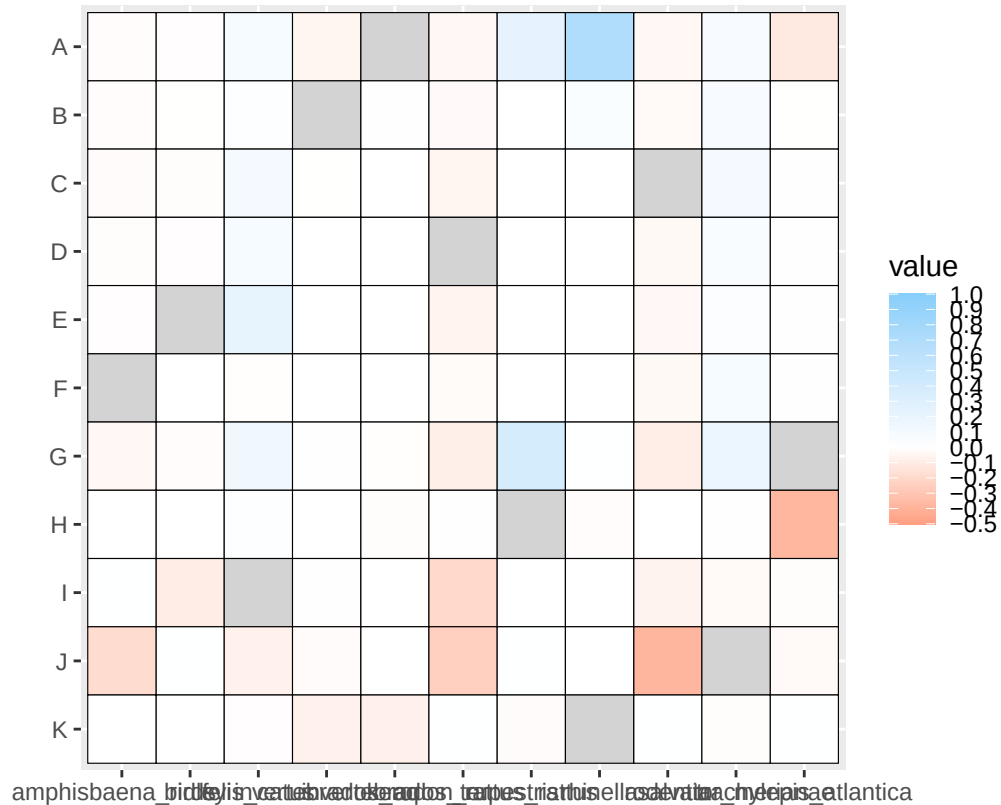
p1 <- ggplot(
  df,
  aes(
    x = x,
    y = ordered(y, levels = rev(sort(unique(y)))),
    fill = value
  )
) +
  geom_tile(color = "black") +
  scale_fill_gradient2(
    low = "#FF0000",
    mid = "#FFFFFF",
    high = "#0000FF"
  )
```

```

high = "#87CEFA",
n.breaks = 15,
limits = c(-0.5, 1),
na.value = "lightgrey"
) +
coord_fixed() +
xlab("") +
ylab("")

```

p1



Histograms of positive and negative effects

```

df <- data.frame(value = as.numeric(PT))

a <- ggplot(df, aes(x = value)) +
  geom_histogram(
    fill = "darkgrey",
    color = "black"
  ) +
  ylim(c(0, 80)) +
  theme_classic() +

```

```

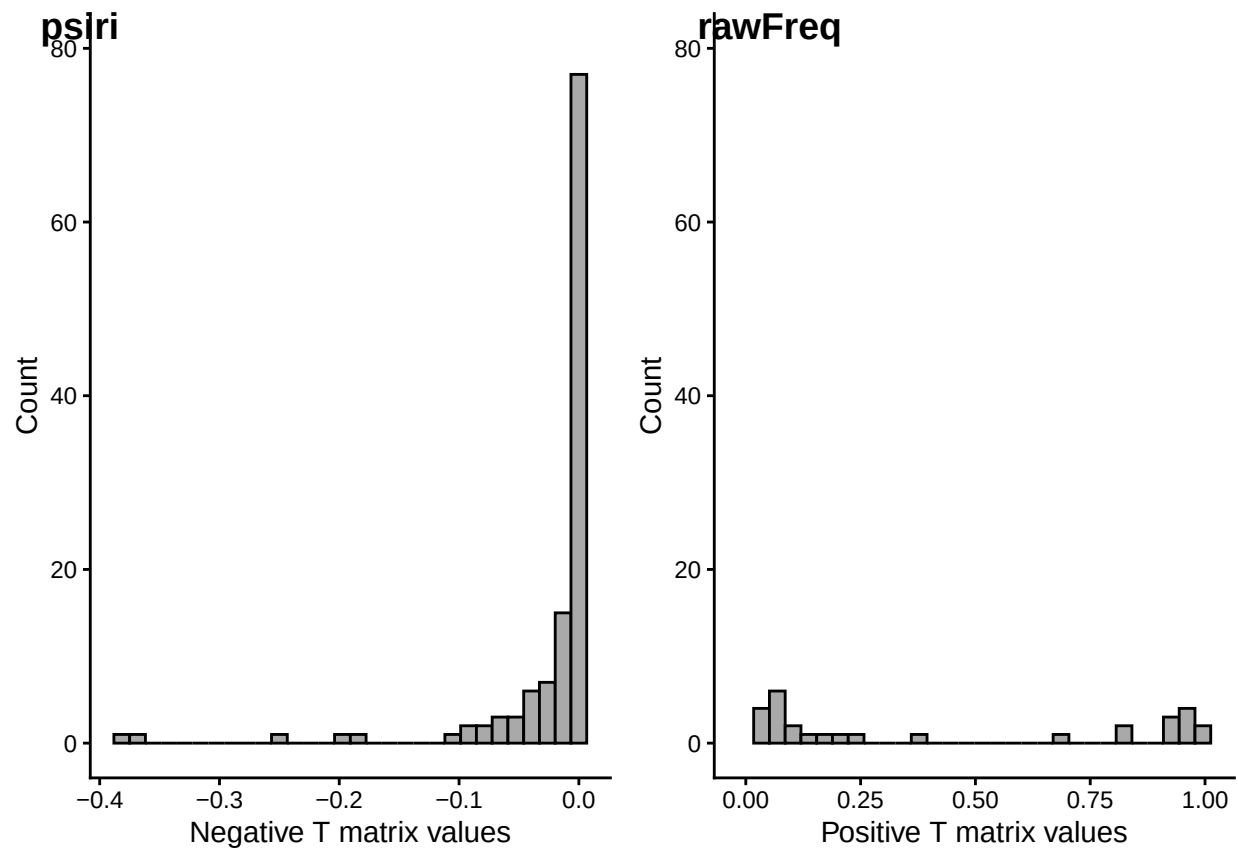
xlab("Positive T matrix values") +
ylab("Count")

df <- data.frame(value = as.numeric(NT))

b <- ggplot(df, aes(x = value)) +
  geom_histogram(
    fill = "darkgrey",
    color = "black"
  ) +
  ylim(c(0, 80)) +
  theme_classic() +
  xlab("Negative T matrix values") +
  ylab("Count")

plot_grid(b, a, labels = c("psiri", "rawFreq"))

```



Sensitivity analysis

Species rankings were recomputed after randomly assigning interaction dependency values (R) 1000 times.

```

r.neg <- numeric(1000)
r.pos <- numeric(1000)

```



```

for(i in seq_len(1000)){

  R.temp <- runif(nrow(psiri))

  Tm.temp <- indirect(M, R.temp)$Tmat

  NT.temp <- (Tm.temp < 0) * Tm.temp
  PT.temp <- (Tm.temp > 0) * Tm.temp

  NTout.temp <- colSums(NT.temp)
  PTout.temp <- colSums(PT.temp)

  r.neg[i] <- cor(
    NTout,
    NTout.temp,
    method = "spearman"
  )

  r.pos[i] <- cor(
    PTout,
    PTout.temp,
    method = "spearman"
  )
}

```

Sensitivity results

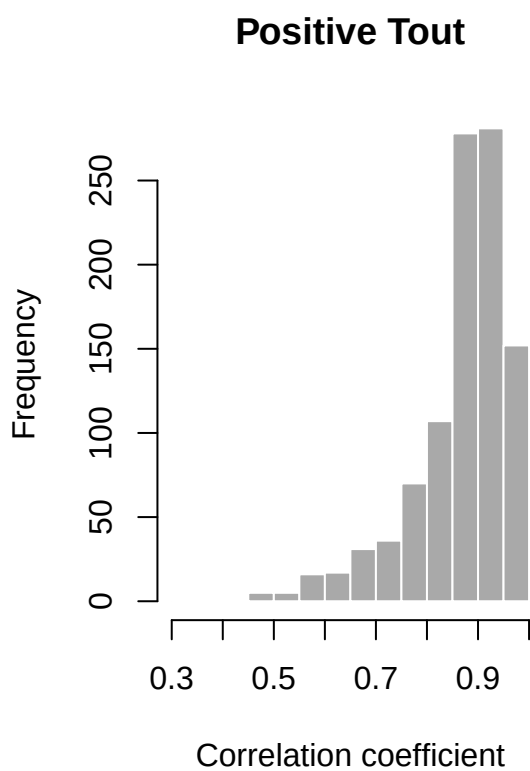
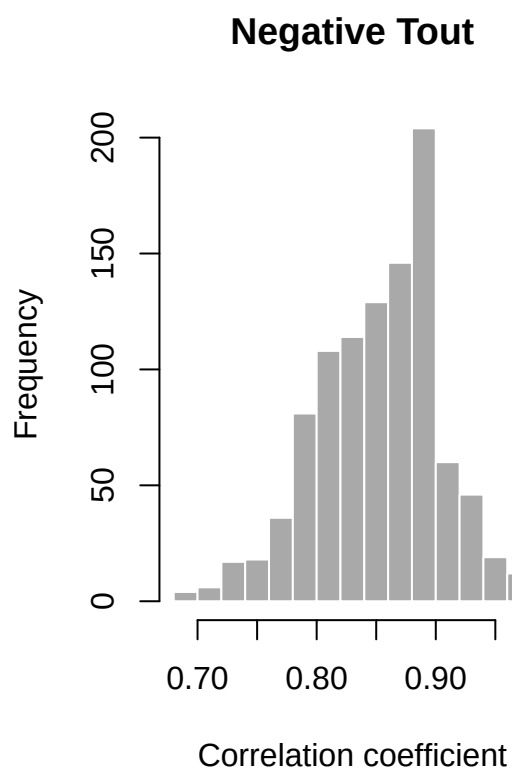
```

par(mfrow = c(1,2))

hist(
  r.neg,
  col = "darkgrey",
  border = "white",
  xlab = "Correlation coefficient",
  main = "Negative Tout"
)

hist(
  r.pos,
  col = "darkgrey",
  border = "white",
  xlab = "Correlation coefficient",
  main = "Positive Tout"
)

```



```
mean(r.neg)
```

```
## [1] 0.8527455
```

```
mean(c(r.neg, r.pos))
```

```
## [1] 0.8608591
```